

William Hao

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EDUCATION

The University of Texas at Austin

May 2027

B.S. Computer Science & B.S. Mathematics – GPA: 3.9/4.0

Relevant Coursework: Computer Architecture, Data Structures, Discrete Math, Stochastic Processes, Linear Algebra, Probability

EXPERIENCE

Citi, Dallas, TX

Jun 2026

Incoming Software Engineering Intern

- Selected for Citi's Technology Summer Analyst (Application Development) program; team placement pending.

Texas Undergraduate Computational Finance, Austin, TX

Jan 2025 – Present

Software Engineer • [GitHub](#)

- Built full-stack React/Flask trading dashboard for 50+ users, automating P&L calculation from raw trade emails.
- Engineered Python NLP pipeline scraping 4000+ NFL transcripts to model 450+ Kalshi football markets.

UT Austin Department of Mathematics, Austin, TX

Jan 2025 – Apr 2025

Undergraduate Researcher • [Presentation](#)

- Applied ergodic theory and Weyl's Equidistribution Theorem to model leading-digit distributions in large datasets.
- Authored 18-page technical paper linking Benford's Law to statistical fraud detection; selected to present to 30+ faculty.

PROJECTS

Automated Hockey Trading Engine

Technologies: Rust (Tokio, axum), WebSockets, Python (SciPy), JavaScript

- Architected Rust trading system executing live orders on Kalshi hockey markets; profitably deployed with \$5k+ returns.
- Modeled win probabilities via Poisson; calibrated against live orderbooks via grid search over 160k+ combinations.
- Structured 7-crate async codebase with continuously pre-computed trade plans for instant execution on game events.

Prediction Market Trading Infrastructure • [GitHub](#)

Technologies: Python (FastAPI, asyncio), WebSockets, SQLite, JavaScript, DigitalOcean

- Built custom live trading dashboard that streams market data and places real orders on Kalshi with <250ms p99 latency.
- Implemented RSA-PSS signing and real-time orderbook synchronization; achieved 100% order placement success rate.
- Deployed Polymarket monitor bot tracking 20+ sharp traders with real-time Telegram alerts on high-conviction positions.

Marvis (HackMIT 2025) • [GitHub](#)

Technologies: Python, TypeScript, MentraOS SDK, Bun, Docker, AWS S3

- Shipped and demoed AR "handyman" assistant in <24h; won 1st Place (Mentra Sponsor Track) among 300+ teams.
- Engineered stateful voice intent parser on MentraOS, enabling hands-free UI navigation with <100ms latency.
- Designed multi-model AI backend (Cerebras, Anthropic) to generate AR instructions from barcode scans in <300ms.
- Developed real-time ingestion pipeline to process live camera inputs and persist session data to AWS S3.

willhao.com • [GitHub](#)

Technologies: TypeScript, Next.js, React, REST APIs, HTML/CSS, Nginx, PM2

- Built full-stack TypeScript portfolio with Next.js/React frontend and custom backend for dynamic data widgets.
- Integrated Spotify OAuth 2.0 and Chess.com APIs with custom endpoints to display live listening history and game stats.
- Achieved 100/100 Google Lighthouse performance score; self-hosted on DigitalOcean behind Nginx with PM2 and SSL.

SKILLS

Programming Languages: Python, Java, C, C++, Rust, TypeScript, JavaScript, HTML/CSS, SQL, Verilog/SV

Technologies: React, Next.js, Node.js, Git, Flask, pandas, NumPy, SciPy, APIs, OAuth 2.0, WebSockets, PostgreSQL, SQLite, Supabase, Jupyter Notebook, PyTorch, LLMs, Docker, Nginx, PM2, Linux, AWS (EC2, S3), DigitalOcean

Honors/Awards: HackMIT 2025 1st Place (Mentra Glass Sponsor Track), USA Computing Olympiad Gold (Java, 2021)

Interests: Competitive chess (USCF 1815; Chess.com 2200), Powerlifting